Hall Ticket Number:

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**MR-22**

**Code: B424CS5**

**MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)**

B.Tech. IV Semester Regular Examinations, July/August-2024

**Software Engineering**

*(Common to CSE, CSB & CSD)*

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

***Part-A (10 x 1 = 10 Marks)***

| Q. No. | Stem of the Question | M | L | CO | PO |
|---|---|---|---|---|---|
| 1. a) | Explain is legacy software. | 1 | 1 | 1 | 1 |
| 1. b) | List the types of software myths. | 1 | 1 | 1 | 2 |
| 1. c) | Explain the kinds of system requirements. | 1 | 2 | 2 | 1 |
| 1. d) | Discuss data flow model. | 1 | 1 | 2 | 2 |
| 1. e) | Justify the importance of refactoring. | 1 | 1 | 3 | 1 |
| 1. f) | Define software architecture with its importance. | 1 | 2 | 3 | 2 |
| 1. g) | Compare verification and validation. | 1 | 3 | 4 | 1 |
| 1. h) | Describe smoke testing. | 1 | 1 | 4 | 2 |
| 1. i) | List out the generic subcategories of predictable risks? | 1 | 2 | 5 | 1 |
| 1. j) | Define risk components? | 1 | 2 | 5 | 2 |

***Part-B (5 x 10 = 50 Marks)***

| Q. No. | Stem of the Question | M | L | CO | PO |
|---|---|---|---|---|---|
| 2. a) | Explain the evolving role of software? | 5 | 1 | 1 | 1 |
| 2. b) | Describe "Software myth"? Discuss on various types of software myths and the true aspects of these myths? | 5 | 1 | 1 | 2 |
| | **OR** | | | | |
| 2. c) | Explain about the evolutionary process models. | 5 | 2 | 1 | 1 |
| 2. d) | Compare and contrast perspective process models and iterative process models. | 5 | 2 | 1 | 2 |
| 3. a) | Write short notes on user requirements. What are requirements? | 5 | 2 | 2 | 1 |
| 3. b) | Compare functional requirements with non-functional requirements. | 5 | 2 | 2 | 2 |
| | **OR** | | | | |
| 3. c) | Explain state machine model with a suitable example. | 5 | 2 | 2 | 1 |
| 3. d) | Demonstrate class hierarchy for library by using interface specification. | 5 | 2 | 2 | 2 |
| 4. a) | Explain a two level process? Why should system design be finished before the detailed design, rather starting the detailed design after the requirements specification? Explain with the help of a suitable example | 5 | 5 | 3 | 1 |
| 4. b) | Explain briefly the following: i) Coupling between the modules, ii) The internal Cohesion of a module. | 5 | 3 | 3 | 2 |
| | **OR** | | | | |
| 4. c) | Describe the way of conducting a component level design. | 5 | 3 | 3 | 1 |
| 4. d) | Explain the guide lines of component level design. | 5 | 6 | 3 | 2 |
| 5. a) | Explain about the importance of test strategies for conventional software. | 5 | 2 | 4 | 1 |
| 5. b) | Compare black box testing with white box testing. | 5 | 2 | 4 | 2 |
| | **OR** | | | | |

**P.T.O.**

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| 5. c) | Describe metrics for source code and for testing. | 5 | 3 | 4 | 1 |
| 5. d) | Demonstrate metrics for analysis model. | 5 | 4 | 4 | 2 |
| 6. a) | Elaborate the concepts of Risk management Reactive vs Proactive Risk strategies. | 5 | 1 | 5 | 1 |
| 6. b) | Explain about Quality concepts. | 5 | 1 | 5 | 2 |
| | **OR** | | | | |
| 6. c) | Describe developing a risk table. | 5 | 4 | 5 | 1 |
| 6. d) | Create a risk table for the project that if you are the project manager for a major software company. you have been asked to lead a team that's developing "next generation "word- processing software. | 5 | 6 | 5 | 2 |

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome